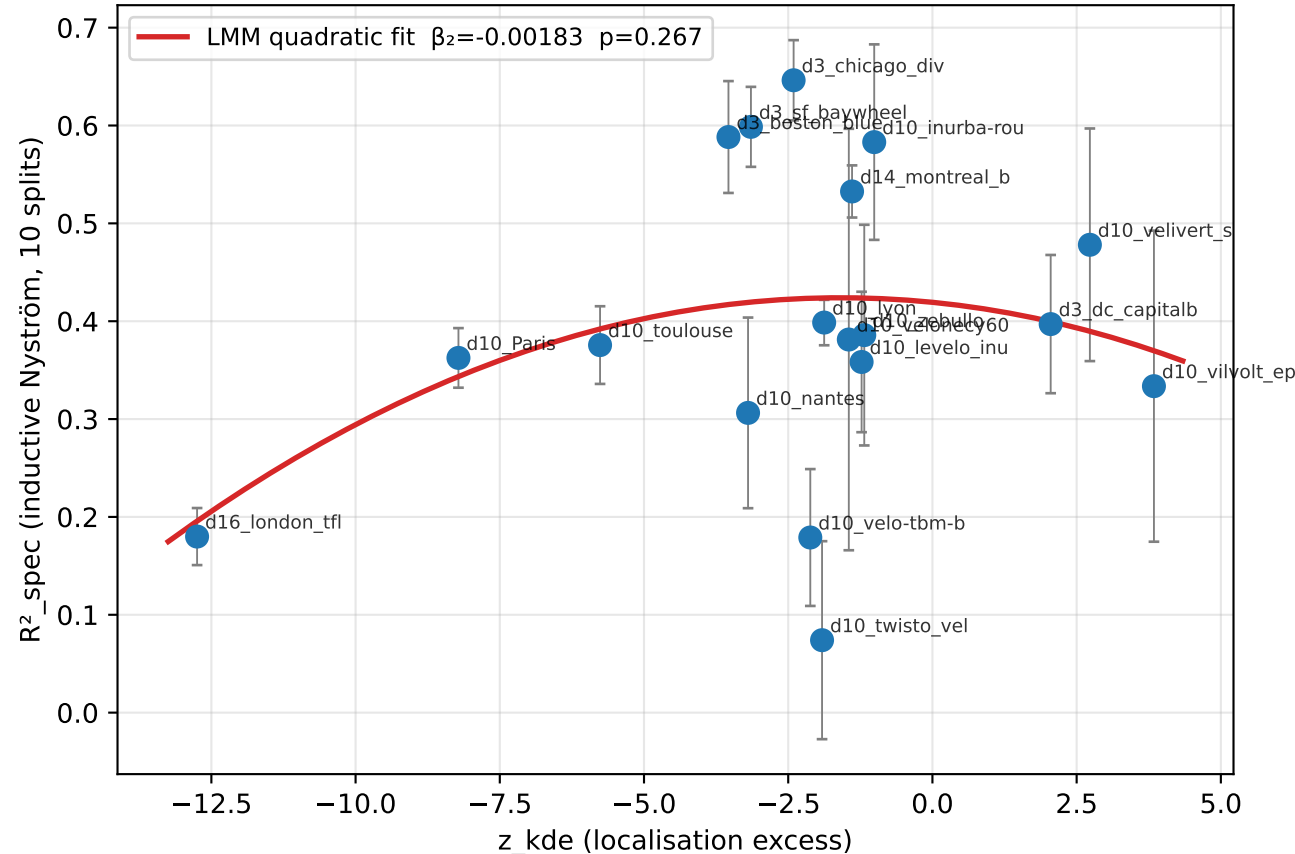


# Goldilocks-zone test of the applicability bound (revised H1)

(a) Goldilocks test on 18 real-demand cities  
strict inductive Nyström,  $\log(y+1)$



## LINEAR MIXED-EFFECTS MODEL

$$R^2_{\text{spec}} \sim z_{\text{kde}} + z_{\text{kde}}^2 + (1|\text{city})$$

$n_{\text{obs}} = 180$

$n_{\text{cities}} = 18$

$\beta_0$  (intercept) = +0.4193

$\beta_1$  ( $z_{\text{kde}}$ ) = -0.00584  $p = 0.7311$

$\beta_2$  ( $z_{\text{kde}}^2$ ) = -0.00183  $p = 0.2672$

Quadratic coefficient sign: ✓ negative

Significant at  $\alpha=0.01$ : ✗ no

$\Delta \log\text{-lik vs linear-only: } -4.872$

Pre-registered H1\*:  $\beta_2 < 0, p \leq 0.01$

VERDICT: NOT SUPPORTED